PhD

Claudia Leticia Gomez Flores

+49 1522 7529 814
claudialeticiaag@gmail.com
https://tipiorgup.github.io/

Computational biophysicist focused on peptide and protein modeling, physics based scoring and machine learning guided design in low data regimes.
Experienced in building automated workflows, databases and software tools that connect simulation, prediction and wet lab testing, and in turning experimental data into design hypotheses.

Research Experience

Postdoc

07/2025 – present

Supervision Dr. Kelvin Anggara (k.anggara@fkf.mpg.de)

Anggara Group, Max Planck Institute for Solid State Research

- Project "**Virgil**: *Model Building for SPM Images of Macro(bio)molecules*": GUI-based workflows for fitting biomolecula models of SPM images.
- First author of **MISO**, a public tool that automates 3D model building from user hypotheses: quaternion based orientation search with geometric and steric scoring, followed by biased force field MD relaxation. Scales from 89 atom glycopeptides to a 2042 atom glycoprotein.
- Unpublished extensions: coarse grained modelling and LLM assisted input generation.
- Developing a machine learning surrogate for fast STM simulation.

Postdoc

03/2024 – 03/2025

Supervision Dr. Christopher V. Synatschke (synatschke@mpip-mainz.mpg.de) and Prof. Tristan Bereau (bereau@uni-heidelberg.de)

Max Planck Institute for Polymer Research

- Project "**SAPs4Tissue**: *Self assembling bioactive peptides for the biomimetic design of functional cell niches in human tissue models*".
- Studied self assembling peptides as intrinsically disordered peptides through coarse grained simulations.
- Used regression models linking sequence and simulation features to experimental aggregation and enhancement of viral infectivity, on a low data regime of 163 known sequences with unnormalized data.
- Benchmarked ProtVec and ProtT5 embeddings for short peptides without PDB background, characterized their limits and augmented them with simulation derived geometry and assembly descriptors.
- Screened coassembly *in silico* to propose new sequence pairs, handed over to experimental collaborators for testing.
- Developed and maintained an internal peptide database linking sequences, simulation observables and prioritized coassembly pairs, bridging computation and wet lab work.

Teaching tasks "Theoretical Statistical Physics" at Heidelberg University

Ph.D. thesis

05/2019 – 10/2022

Advisor Prof. Marcus Elstner (marcus.elstner@kit.edu)

Karlsruhe Institute of Technology

- "**Research Training Group 2450**: *Tailored Scale Bridging Approaches to Computational Nanoscience*".
- Neural network delta correction implemented in DFTB+, with genetic algorithm descriptor optimization, trained on about 10K unique structures. Corrects DFTB/MM to B3LYP and CCSD(T) level at 3 to 4 orders of magnitude lower cost.
- Model transferability to bigger and solvated model systems.
- QM/MM study of thiol disulfide exchange in an immunoglobulin domain under mechanical stress using multiple walker metadynamics.

- Integrated data analysis, machine learning and computational chemistry workflows into an interactive GUI for non expert users.

Teaching tasks "Physical Chemistry Laboratory for Beginners" and "Machine Learning for Chemists" at Karlsruhe Institute of Technology

M.Sc thesis — 08/2017 – 02/2019

Advisor Prof. Jose G. Sampedro Pérez (sampedro@mail.ifisica.uaslp.mx)

Universidad Autónoma de San Luis Potosí

- Performed full atomistic MD simulations and molecular docking to identify and characterize ion and nucleotide binding sites in the N domain of membrane H^+ATPase from *Saccharomyces cerevisiae*.
- Conducted functional and structural analysis using circular dichroism, absorbance and fluorescence spectroscopy, including intrinsic fluorescence quenching and terbium luminescence.

Internship — 02/2018 – 10/2018

Supervision Prof. Edgar A. Cerda Méndez

Universidad Autónoma de San Luis Potosí

- Assembly of an optical arrangement for microcavity characterization at cryogenic temperatures.
- Reflectance, photoreflectance and photoluminescence measurements on different microcavities in real and reciprocal space.

Internship — 05/2017 – 08/2017

Supervision Dr. Bernardo Yáñez Soto

Universidad Autónoma de San Luis Potosí

- Characterization of glass surfaces by silanization and ionic coating.
- Measurement of dynamic contact angles between liquid interfaces over the characterized surfaces.

B.Sc thesis — 06/2016 – 02/2017

Advisor Prof. Jose G. Sampedro Pérez (sampedro@mail.ifisica.uaslp.mx)

Universidad Autónoma de San Luis Potosí

- Expression and purification of the recombinant N domain of the H^+ATPase of *S. cerevisiae*.
- Steady state fluorescence experiments for determining the position of nucleotide binding sites.

Education

Ph.D. Natural Sciences *magna cum laude* — 2019 – 2022

Karlsruhe Institute of Technology, Germany. Specialization: Physical Chemistry

M.Sc Interdisciplinary Sciences — 2017 – 2019

Universidad Autónoma de San Luis Potosí, México. Specialization: Biochemistry

B.Sc Biophysics — 2013 – 2016

Universidad Autónoma de San Luis Potosí, México. Earlier: B.Sc Physics, UNAM (2011 – 2012)

Additional Notes

Maternity Leave **12/2022 – 01/2024**

Publications

Google Scholar profile

Electrostatic interactions contribute to the control of intramolecular thiol–disulfide isomerization in a protein. D Maag, M Putzu, **CL Gómez Flores** *et al*. Physical Chemistry Chemical Physics (2021)

Accurate Free Energies for Complex Condensed Phase Reactions Using an Artificial Neural Network Corrected DFTB/MM Methodology. **CL Gómez Flores**, D Maag (equal contribution), M Kansari *et al*. Journal of Chemical Theory and Computation (2022)

Identification of Calcium Binding Sites in the Nucleotide Binding Domain of the Plasma Membrane H+ATPase from *Saccharomyces cerevisiae*. **CL Gómez Flores**, JG Sampedro Perez. ChemPhysChem (2026)

Automating Model Building for SPM Images of Biomolecules Using MISO. **CL Gómez Flores**, K Anggara. Journal of Chemical Information and Modeling (2026)

Skills

Computational methods

- Full atom and coarse grained molecular dynamics
- Enhanced sampling and free energy (metadynamics)
- QM/MM, quantum chemical calculations
- Molecular docking
- Scientific GUI development
- Workflows, databases and data analysis pipelines
- Data visualization
- HPC and GPU: SLURM, LoadLeveler, qsub queues

Machine learning methods

- Protein language models (ProtVec, ProtT5)
- SVM, regression, classification
- Neural networks, delta learning, surrogate models
- Molecular descriptors (ACSF, SLATM)
- Learning in low data regimes, data mining
- LLM assisted tooling

Experimental methods

- Protein expression
- Protein purification
- Circular dichroism
- Fluorescence spectroscopy
- Absorbance spectroscopy
- Optical setups